

THE TWENTY-FIFTH WILLIAM LOWELL PUTNAM MATHEMATICAL COMPETITION

December 5, 1964

Morning Session

1. Given a set of 6 points in the plane, prove that the ratio of the longest distance between any pair to the shortest is at least $\sqrt{3}$. (page 586)

2. Find all continuous positive functions $f(x)$, for $0 \leq x \leq 1$, such that

$$\int_0^1 f(x) dx = 1$$

$$\int_0^1 f(x)x dx = \alpha$$

$$\int_0^1 f(x)x^2 dx = \alpha^2$$

where α is a given real number.

(page 587)

3. Let $P_1, P_2, \dots$ be a sequence of distinct points which is dense in the interval $(0, 1)$. The points $P_1, P_2, \dots, P_{n-1}$ decompose the interval into n parts, and P_n decomposes one of these into two parts. Let a_n and b_n be the lengths of these two intervals. Prove that

$$\sum_{n=1}^{\infty} a_n b_n (a_n + b_n) = 1/3.$$

(A sequence of points in an interval is said to be dense when every subinterval contains at least one point of the sequence.) (page 588)

4. Let p_n ($n = 1, 2, \dots$) be a bounded sequence of integers which satisfies the recursion

$$p_n = \frac{p_{n-1} + p_{n-2} + p_{n-3}p_{n-4}}{p_{n-1}p_{n-2} + p_{n-3} + p_{n-4}}.$$

Show that the sequence eventually becomes periodic.

(page 589)

5. Prove that there is a constant K such that the following inequality holds for any sequence of positive numbers $a_1, a_2, a_3, \dots$:

$$\sum_{n=1}^{\infty} \frac{n}{a_1 + a_2 + \dots + a_n} \leq K \sum_{n=1}^{\infty} \frac{1}{a_n}. \quad (\text{page 589})$$

6. Let S be a finite subset of a straight line. Say that S has the repeated distance property when every value of the distance between pairs of points of S (except for the longest) occurs at least twice. Show that if S has the repeated distance property then the ratio of any two distances between two points of S is a rational number. (page 592)

Afternoon Session

1. Let u_k ($k = 1, 2, \dots$) be a sequence of integers, and let V_n be the number of those which are less than or equal to n . Show that if

$$\sum_{k=1}^{\infty} 1/u_k < \infty,$$

then

$$\lim_{n \rightarrow \infty} V_n/n = 0. \quad (\text{page 594})$$

2. Let S be a set of $n > 0$ elements, and let $A_1, A_2, \dots, A_k$ be a family of distinct subsets, with the property that any two of these subsets meet. Assume that no other subset of S meets all of the A_i .

Prove that $k = 2^{n-1}$. (page 595)

3. Let $f(x)$ be a real continuous function defined for all real x . Assume that for every $\epsilon > 0$

$$\lim_{n \rightarrow \infty} f(n\epsilon) = 0, \quad (\text{where } n \text{ is a positive integer}).$$

Prove that

$$\lim_{x \rightarrow \infty} f(x) = 0. \quad (\text{page 595})$$

4. Into how many regions do n great circles (no three concurrent) decompose the surface of the sphere on which they lie? (page 597)

5. Let u_n ($n = 1, 2, 3, \dots$) denote the least common multiple of the first n

terms of a strictly increasing sequence of positive integers (for example, the sequence 1, 2, 3, 4, 5, 6, 10, 12, ...). Prove that the series

$$\sum_{n=1}^{\infty} 1/u_n$$

is convergent.

(page 598)

6. Show that the unit disk in the plane cannot be partitioned into two disjoint congruent subsets.

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